

APPLICATION FOR ESTABLISHMENT OF GREATER MASAKA REGIONAL CENTRE OF EXPERTISE

BY

UGANDA MARTYRS UNIVERSITY

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Application for the establishment of a Regional Centre of Expertise in Greater Masaka region.

Presented by, Uganda Martyrs University.

Name: The name of the regional center of expertise is proposed as:

Greater Masaka Regional Centre of Expertise (GMARCE)

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1.0 Introduction

1.1 Location and Physical Features

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Uganda lies astride the equator in the eastern region of Africa and is located on the raised part of the African plateau. She is a landlocked country bordering Kenya to the east, South Sudan to the north, Tanzania to the south, the Democratic Republic of Congo to the west and Rwanda in

the southwest. She lies within the latitude of 4° 12' N and 1° 29' S, longitude of 29° 34' E and 35° 0' W. The total surface area is 241,551 km² of which 199,807 km² fall under land and 41,743 km² under water and swamps. The annual temperature and rainfall range between 12°C to 32°C and 800 to 1,700 mm, respectively. Uganda's landscape has rich diversity of relief features with great contrasts bearing significant influence on the physical and human environment.

on the physical, biological and human environments. For example Uganda has rich and diverse climatic conditions, ecosystems and other natural resources, which include fresh water lakes, rivers, wetlands, fisheries, forests, wildlife, minerals, and soils that support different agricultural systems. Gifted by nature, the country was once referred to by Sir Winston Churchill as the 'Pearl of Africa'.

Uganda's total population has grown to 34.5 million, with the proportion of females and males being the same at 17.3 million each. This is up from 33.8 million Ugandans in 2010. According to the latest state of the world population report by the UN Fund for Population Activities (UNFPA), Uganda's annual population growth rate has fallen from 3.3% in 2010 to 3.1% in 2011.

Masaka lies approximately 140km (87 mi) to the southwest of Kampala, on the highway to Mbarara. The town lies very close to the Equator, as can be seen from the latitude values. It was Uganda's second-biggest town for a long time. Today this status has changed, as the city was largely destroyed in the Uganda-Tanzania War of 1979 and again in 1981-1986 civil war that removed Milton Obote from power for the last time.

2.0 Environmental/Natural Resources

2.1 Water

The water resources of the Greater Masaka Basin comprises of surface water resources (Lake Victoria, satellite lakes, rivers, dams and ponds), ground water resources, and rainwater. Lake Victoria is the main surface water body in the Basin and hence defines the operational framework for the Basin activities. There are a number of other smaller lakes of varying depths, major inflowing rivers, dams and ponds, extensive riverine and lacustrine swamps. Direct rainfall over the Lake contributes by far the greatest (82 %) source of water to Lake Victoria and the rest comes via major inflowing rivers. Ground water resources are regarded as potential sources of water for domestic and commercial purposes. However, exploitation of ground water resources is still limited. This resource supports most of the productive sectors in the Basin including agriculture (both irrigated and rain fed), wildlife, tourism and fisheries among others.

2.2 Fish

The fishery resources from the Lake are, directly or indirectly, a source of livelihood to the people engaged in subsistence, artisanal and commercial fishing. Fishing is next to agriculture in terms of economic activity in the basin. However, there is decline in fish diversity and catch per effort due to over fishing or excessive fishing effort and use of destructive fishing gears, destruction of fish breeding and nursery habitats, introduction of alien fish species (notably the Nile perch), water pollution and rapid increase in demand for fish. The consequences of this have been poor fish yields, loss of unemployment opportunities and livelihoods, malnutrition, increase in national as well as regional and community conflicts. In respect to fisheries resources decline (an indication of biodiversity loss in the Basin), the most important concern is the evident loss in fish species diversity over the last 50 years. Within this period, an estimated over 200 fish species have disappeared from the original 400-500 species known in Lake Victoria.

2.3 Wetlands

Wetlands that fringe the shores of Lake Victoria and its river systems, are among the most productive ecosystems in the Basin. They include a variety of swamps, marshes and seasonally inundated habitats, which are a home to various plants (e.g. sedges, *Cyperus papyrus*, date palm and grasses) and animals (e.g. hippopotamus, sitatunga, crocodiles and snakes). The greatest concentration of wetlands is in Uganda, which has the highest diversity of aquatic and semi-aquatic plants (NBI, 2001).

The wetlands of lake Victoria also act as buffer for the water systems by removing pollutants, nutrients and sediment entering from rivers feeding into the Lake, and from near-shore overland runoffs. They also act as a water regulator by sustaining water supply during the dry season. The water that moves slowly through wetlands is received downstream much later in the season, maintaining the flow of rivers and the water in the Lake.

Pollution hotspots in major urban centers and industries discharge their effluents into sheltered bays and wetlands. This affects wetland quality and hence their beneficial use by the riparian communities. Some wetlands have lost their stripping capacity and this contributes to the nutrient loading of the entire lake.

Wetlands support a number of socioeconomic activities, including crop, flower and horticultural farming, livestock keeping, fishing, craft making; grass, fuelwood, timber and medicinal plants collection, sand mining, clay and stone extraction, hunting and tourism. However, wetland degradation threatens their continued beneficial use. A number of wetlands have been protected through global efforts declaring them protected **Ramsar sites** such as Wetland, Mabamba, Sango Bay, Nabajjuzi wetland, Lake Mburo and Lutembe Bay within the Greater Masaka.

Wetland destruction (extent and functionalities) is location specific but with regional consequences. The major primary sources of degradation are agricultural expansion (search for food security and sufficiency), least cost waste disposal practice, over-extraction of wetland products to meet household needs (subsistence and/or cash), and infrastructure development. The continued wetland encroachment and discharge of waste beyond the assimilative capacity of these buffer zones will impair their functionalities resulting into leakage of the waste or excessive nutrients into the region.

2.4 Forests

Forests offer **the main terrestrial system that supports the Greater Masaka basin**, acting as water regulator and purifier and habitat for different flora and fauna. However, the current trends have seen increased forests degradation due to increased demand and utilization of forestry resources, conversion of forest land to other land uses (agriculture, settlements), and climate change effects. Major water towers in the Basin have been degazetted to pave way for agricultural and other development needs. There is increased loss of indigenous trees species and this has been replaced by exotic trees.

As a direct consequence of forest degradation, the basin continues to experience flash floods and reduced base flows which are common benefits associated with forest cover. Similarly, reduced lake inflows are linked directly to catchments destruction.

3.0 Social and Economic

3.1 Population

Approximately, over 5 million people live in the Greater Masaka region, of which 65% of the population is under 25 years which implies a high dependency level. The majority of the population is dependent on natural resources and small holdings with approximately 75% subsistence on 1 ha or less.

Over the last two decades, the Basin has been ravaged with HIV/AIDS. HIV/AIDS prevalence varies from one district to another. The prevalence varies around 4% in the Greater Masaka region. With concerted government efforts, HIV/AIDS prevalence has stabilized or is declining. Average life expectancy is relatively low, with an average of 44.6 years for males and 47.8 years for females. Poverty levels are rampant and it is estimated that over 50% of the total population in the Basin lives under the poverty line.

There is a continuous threat of outbreaks of waterborne diseases in the Basin. Available data shows numerous limited outbreaks of diseases caused by exposure to or consumption of poor quality water containing pathogenic bacteria that are responsible for transmitting various contagious diseases.

3.2 Energy

Generally, wood fuel supplies over 90% of the energy requirement in the Basin. Other potential energy sources such as solar, geothermal, and wind energy are not significantly developed. Currently the region has a deficit in sustainable wood supply. Consequently it is scarce and requires more time to collect it.

Wind power is used mainly to pump water in few parts of the region and to propel sailboats in the lake. Hot springs are present in parts of the region, but these have not been exploited for energy. Similarly, conversion of solar power to electric energy is limited.

Hydro-power is the main source of energy for urban utilization and industrial use, however, declining water levels of Lake Victoria has for the last 5 years adversely affected hydro-power generation at Jinja.

Due to hampered expansion of hydro-power generation and associated poor infrastructure, electricity supply is still very low within the region. Energy development has been looked at improving the welfare of women and girls since they bear the challenges associated with its underdevelopment most. 90% of the people rely on Biomass for their energy sources.

3.3 Pollution

Lakes water is becoming more eutrophic from nutrients inputs from the atmospheric deposition, nutrient runoff from agriculture areas and, urban industrial effluents. Persistent pollutants is another class of pollutants that impacts on the water quality in the basin featuring the following major POPs: DDT, Aldrin, Chlordane, Dieldrin, Dioxin, Endrin, Furans, Heptachlor, Hexachloro Benzene, Mirex, and Toxaphene.

Consequences of reduced water quality are increased use of poor quality water, increased water borne diseases, increased costs for water treatment, frequent fish kills, poor fish catches, and over-all decline in water resources productivity.

4.0 Climate

The large size of the region makes it vital for weather and climate modulation. The seasonal wind patterns influence the hydrological process in the region. The convergence of winds over the lake, twice a year for instance, accounts for heavy rainfall amounts of the region (Lake Victoria Environmental Management Project, 2002).

5.0 Environmental Governance

The governance of the environmental resources in all the region takes its roots from the laws and policies put in place by the colonial governments. These policies and laws focused on utilization rather than conservation and as such, they were not comprehensive enough to respond to the environmental problems. However, from the early 1990's Uganda have

undertaken marked reforms and developments of the environmental policies and laws including creation and strengthening of institutions for the environmental management.

6.0 Natural Resources and Economic Potential

Most of the people in this region are farmers with both cattle and other crops. Greater Masaka used to be the highest producer of the indigenous banana food (locally referred to as Matooke). Other economic activities include; Meat & Fish Processing, Production of Beverages, Furniture Manufacturing, Manufacture of Footwear, Manufacture of Baked Products, and Glass Manufacturing. One of the main economic activities in Masaka is 'gathering grasshoppers' that can potentially earn a collector over US \$20,000 (UGX: 50 million) annually. (US \$1 = UGX: 2,500/=)

Agriculture in this region shows productivity decline as the rapidly growing population overexploits the land resources (soil and vegetation cover) through impacts of deforestation of Kalangala forests, Mpanga forests, and encroachment on swamps for agricultural expansion. Climate change has also been witnessed where presently shorter rainy periods with longer dry spells have been realized. This has threatened food safety and security, poverty and famine. Animal husbandry is characterized by keeping of the indigenous varieties of cattle, goats and piggery. The region is richly endowed with tourist attractions which include L. Victoria, R. Katonga which also support fishing as an economic activity.

7.0 Major Challenges to Sustainable Development in Greater Masaka Region

Greater Masaka region is faced with a multitude of challenges. These were identified through various information sources such as research reports, projects reports workshops, conference reports, policy documents of Uganda government, NGOs, local government (district) state of environment reports, field experiences narrated by key stakeholders and media. The Challenges were comprehensively discussed during stakeholder meeting at Uganda Martyrs University (11th July 2013), leading to the identification and prioritization of key challenges which are broadly categorized into the themes of ecological, economic, social, health and educational under the following headings:

a) Deforestation

Generally, the livelihoods of the local community in this region are linked to natural resource exploitation and ecosystem services. Thus the environmental concerns associated with agriculture relate mainly to the sustainability of the resource base for agricultural production (e.g. soil quality), protection of biodiversity and habitat and environmental/ecosystem services of resources influenced by agricultural land use (e.g. carbon sequestration). Agricultural expansion, intensification and de-vegetation are the leading causes of species loss and

depletion of natural vegetation. Like other parts of Uganda, there is a special concern over increasing land degradation in Greater Masaka region. The increased demand for agricultural land, wood fuel and building material, has led to draining of wetlands, deforestation, pollution and land degradation. Need for more fertile and virgin land has resulted into encroachment on fragile environments components such as wetlands and forest reserves for Greater Masaka region where the oil palm plantation has claimed the biggest percentage of Kalangala virgin forests and the nearby swamp vegetation. As a result, degradation of soil and vegetation cover due to application of synthetic agro-chemicals and other inputs to achieve instant agricultural productivity and returns has been realized and accelerated climate change, poverty and threats to food security. Thus loss of soil fertility with the consequent reduction in agricultural productivity is also one of the major concerns in the region. This leads to gaps in food production against the increasing human population. Environmental degradation especially the loss of vegetation cover is one of the major constraints to improved livelihoods of the community in the area. This has resulted in lack of source of biomass for household energy needs.

b) Degradation of Land, Water and Catchment Areas:

With land as the main capital resource for people in Uganda, it constitutes the main support for agriculture on which the country depends. Uganda's economy basically depends on natural resources like land, forestry, and water resources as an engine for economic growth and as a basis for rural livelihoods. Inappropriate land and water management is a major problem constraining the development of agriculture in Africa. Greater Masaka region in Uganda suffers from degraded soils, which are less productive because of poor management and inappropriate farming practices. Intensive tilling of the land, continuous growing of the same crops and inadequate measures to restore organic matter has accelerated nutrients loss and is the basis for soil life and productivity. Rainwater fails to seep into compacted or crusted soil; so much of it runs off, carrying with it the valuable topsoil with substantial losses of precious water and plant nutrients. Soil nutrient depletion is thus one of the major constraints to food security and economic development in Masaka region and Uganda. Land degradation has far-reaching implications on livelihoods, ranging from low agricultural productivity (posing a risk on food security, poor nutrition and health) to poverty (resulting in a failure to meet basic social needs such as food, shelter and education).

In addition, encroachment on water catchment areas within Greater Masaka has provided the only alternative to the exhausted and infertile soils. Swamps have been reclaimed, river banks and lake shores and with poor methods of agriculture like synthetic chemical and fertilizer application which when it rains, the toxins have been washed to the fresh waters of nearby rivers and L. Victoria. The result of this has been water pollution and toxin evaporation resulting to climate change. With the global trend of industrialization, lake shores of Victoria have attracted a number of fish processing industries, palm oil processing industries, among others whose waste is dumped in the waters of Victoria and have also accelerated water degradation in addition to the water hyacinth.

c) Waste Management:

In Uganda human population is growing very rapidly. For the Greater Masaka Region especially in urban centres, the greatest challenge is the menace of garbage or waste while in rural areas the challenge is sanitation and hygiene. Urban waste includes both organic and inorganic materials produced by households, commercial, institutional and industrial activities. The volume of solid waste generated in urban centres in the region has been increasing mainly as a result of growing urban population, increased concentration of industries, changing consumption habits of residents and inadequate finance and facilities to undertake waste collection and disposal. Non-biodegradable materials particularly polythene bags and plastics constitute a menace to the environment and health. Waste disposal in urban areas has been a real challenge in comparison to rural Uganda where waste is mainly organic and can be disposed in open places, gardens and open pits with less health hazard. Despite the drastic increase in solid waste generation, there has not been a proportional improvement in investment by local governments in solid waste management and therefore leaving a lot of waste indiscriminately disposed which poses health hazard in the urban population in the region.

d) Land

In Uganda, men owned land and they took full responsibility of maintenance, sustainability and development. Acquisition was mainly through inheritance and to some extent monetary. However with the changing political economy, any one can have land as long as they can afford. However, with the changing trend towards development, many theories and approaches that have been employed by developers and owners have threatened sustainability. Land fragmentation too has threatened sustainability for example conventional means of production that limits control of non-compliance towards it.

Conserving and promoting indigenous knowledge is an important and valuable input in the promotion of sustainable development initiatives and management of such resources like land. Over centuries of time, the local community lived in harmony with the environment and they utilized resources without impairing nature's capacity to regenerate them. Their ways of living were sustainable and indigenous knowledge shaped their values and attitudes about resource extraction and utilization. This scenario has changed with the increased population in Greater Masaka region with significant impact on land, together with other resources such as forest and water resources. The community abandoned indigenous and cultural ways of preserving these resources but instead has adopted some disastrous ways of encroachment in the guise of restoring and maintaining them. An example of this scenario is the Kalangala natural forest which has been replaced by the oil palm plantation which has contributed significantly to biodiversity loss and climate change in the region. It is thus necessary to document and preserve such knowledge that has sustained the community over a very long period and for the future generations.

e) Climate Change:

The African Initiative on Climate Change (CIGI, 2007) identified Uganda as one of the most

vulnerable countries in the world to climatic change impacts because the economy and welfare of the population are intricately linked to the natural ecosystem and environment. Currently, 88 percent of Ugandans live in rural areas where land, forestry and water resources are central to their livelihoods. Any negative impact resulting from climate change is likely to negatively affect the communities. The current average temperatures in Uganda are expected to increase by between 0.7°C and 1.5°C by 2020, and rainfall patterns are changing dramatically (Government of Uganda, 2009). This is already severely affecting livelihoods and the ability of populations to cope. According to the State of Environment Report for Uganda (NEMA, 2008) climate change is already affecting food security in the country through reduced production of major food crops, as a result of the increased occurrence of droughts (for longer periods than before), floods, soil erosion and landslides (NEMA, 2008). Increased water stresses are likely to result in declining agricultural productivity, a reduction in yields from rain-fed agriculture and loss of livestock. Already, climate change is adversely affecting the suitability of areas to maintain livestock and grow food and cash crops such as beans, cassava, maize, coffee and plantain (Oxfam, 2008). Most areas in Masaka region have become food insecure. These areas are characterized by reduced rainfall distribution and reliability as well as reduced soil fertility.

f) Energy

Masaka region is under huge threat of resource depletion and degradation due to the fact that people within this region depend largely on wood fuel as the cheap available source of energy. Other energy sources, for example Hydro, biogas solar are too expensive for them to afford. Poverty, being the mainly characteristic of people in Greater Masaka region, many cannot afford the alternative sources. This has therefore brought about increased encroachment at both household and institutional levels of the available resources. However research has proved that several alternatives that are also environmentally friendly are also cheaper for instance bio fuel and solar.

g) Socio-Economic Challenges

With the crippling of main economic activities in Greater Masaka region which were supported by coffee production and also serving as a food basket for the country, the proportion of population living below the poverty line has been reported to increase due to the decline in production of the above activities. Despite the previous economic recovery period registering GDP at 6%, poverty is still a big challenge and is particularly worse in the rural areas of the Masaka region. Unemployment is high in the region and this is exacerbated by rural urban migration that has led to mushrooming of slums with more youths engaging in motorcycle business (Boda-boda) after selling much of their inherited land in villages and grasshopper business (Nsenene). Economic development has partly been hampered by the low investment and saving culture.

8.0 Process of RCE Development

Greater Masaka Regional Centre of Expertise/Excellence (GMARCE) was conceived on the realization of the need to create a forum through which Universities, Local Governments, Higher Institutions, Primary & Secondary schools, NGOs, CBOs, Youth, Communities within the entire region, through their organized groupings, could collectively be involved in Education for sustainable development (ESD) in Formal and Non Formal Education and initiatives directed towards improving the socio-economic livelihoods of the people and while at the same time improving on the environmental and ecological aspects. It is a regional network of Higher Education Institutions and grassroots organizations, which exist to strengthen regional participation in Education for Sustainable Development through research, advocacy, lobbying and information sharing. It is an ESD stakeholder network whose membership is drawn from the regional Districts.

The districts forming Greater Masaka region are within the Lake Victoria Basin and support approximately over 5 million people and the social, economic, environment and ecological values. The basin is well endowed including rich social capital, rich agricultural soils, abundant water resources, minerals, fisheries, wetlands, forest reserves, and wildlife. GMARCE will promote a cooperative approach to sustainable development through collaborative research, policy analysis, capacity building and community-based initiatives. The development of GMARCE has evolved through several initiatives that include:

Meeting at Uganda Martyrs University (11th July 2013)

This meeting brought together a multi-sectoral group of over 40 participants representing local governments from the Greater Masaka region, civil society, the private sector, primary and secondary schools, and higher education institutions (including Uganda Martyrs University and Muteesa I Royal University). The meeting was facilitated by Uganda Martyrs University's Faculty of Education.

The main objectives of the meeting were to:

1. Introduce the concept of Regional Centres of Expertise (RCE) and Education for Sustainable Development (ESD).
2. Brainstorm and identify the key sustainable development challenges facing the Greater Masaka region.
3. Establish a preliminary framework and working group to spearhead the application process for the establishment of GMARCE.

Participants enthusiastically embraced the RCE concept, noting that it provides a unique platform to bridge the gap between academic research and practical community needs. A steering committee was formed, chaired by Uganda Martyrs University, to compile information and draft the formal application.

9.0 Vision, Mission and Objectives of GMARCE

9.1 Vision

A sustainable, resilient, and empowered Greater Masaka region where communities thrive in harmony with a well-managed and healthy ecosystem.

9.2 Mission

To foster, facilitate, and integrate Education for Sustainable Development (ESD) into all levels of formal, non-formal, and informal education, training, and public awareness within the Greater Masaka region, leveraging multi-stakeholder partnerships to address critical local environmental, social, and economic challenges.

9.3 Core Objectives

- **Promote Networking and Collaboration:** Establish a strong, dynamic network linking higher education institutions, research bodies, local governments, non-governmental organizations (NGOs), community-based organizations (CBOs), schools, and the private sector to share best practices, resources, and knowledge on ESD.
- **Capacity Building and Training:** Enhance the capacity of educators, community leaders, local government officials, and youth in integrating sustainable development principles into school curricula, community development plans, and daily practices.
- **Research and Innovation:** Conduct and promote action-oriented, interdisciplinary research focusing on local sustainability challenges—such as soil degradation, deforestation, waste management, and climate change—and develop innovative, practical solutions suited for the region.
- **Community Empowerment and Livelihoods:** Support and scale up grassroots initiatives that improve local livelihoods, enhance food security, and promote eco-friendly economic activities (e.g., sustainable agriculture, clean energy adoption, and waste recycling).
- **Public Awareness and Advocacy:** Raise public awareness on pressing environmental and social issues through targeted media campaigns, community dialogues, and environmental clubs in schools, encouraging behavioral change towards resource conservation.

10.0 Governance and Institutional Framework

To ensure effective coordination, transparency, and long-term sustainability, GMARCE will operate under a clear governance structure comprising the following organs:

10.1 The Advisory Board

The Advisory Board will serve as the supreme oversight body of the RCE. It will consist of high-level representatives from key stakeholder groups, including:

- The Vice Chancellor of Uganda Martyrs University (or designated representative).
- District Chairpersons (LC V) or Chief Administrative Officers (CAOs) from the

- participating districts.
- Regional representatives from the National Environment Management Authority (NEMA) and the Ministry of Education and Sports.
- Representatives from prominent regional NGOs and the private sector.

The primary roles of the Advisory Board will include providing strategic direction, approving annual work plans and budgets, mobilizing financial and technical resources, and monitoring overall progress.

10.2 The Secretariat (Host Institution)

Uganda Martyrs University (UMU) will serve as the official host institution and headquarters for the GMARCE Secretariat. UMU will provide the necessary physical infrastructure, including office space, communication facilities, and administrative support. The Secretariat will be headed by an RCE Coordinator, supported by a dedicated team of technical and administrative personnel.

The Secretariat's responsibilities will include day-to-day coordination of network activities, managing communications, organizing stakeholder meetings, maintaining documentation, and preparing progress reports for the global RCE network and local partners.

10.3 Technical Working Groups (Thematic Committees)

To address the prioritized challenges effectively, GMARCE will establish specialized Technical Working Groups organized around core thematic areas. These groups will comprise experts, practitioners, and community representatives passionate about specific sectors:

1. **Ecology and Natural Resource Management:** Focusing on reforestation, wetland restoration, soil conservation, and water catchment protection.
2. **Sustainable Agriculture and Food Security:** Focusing on promoting organic farming methods, indigenous knowledge conservation, agroforestry, and climate-resilient crop production.
3. **Energy and Appropriate Technology:** Focusing on the research, development, and promotion of affordable renewable energy options (biogas, solar power, fuel-efficient stoves).
4. **Health, Sanitation and Waste Management:** Focusing on urban solid waste management, rural hygiene campaigns, and tackling waterborne diseases.
5. **Curriculum and Pedagogy (Formal Education):** Partnering with primary, secondary, and tertiary institutions to integrate ESD concepts into active teaching and learning processes.

11.0 Stakeholders and Partnership Network

The success of GMARCE relies on its inclusive, multi-stakeholder model. The network brings together diverse actors, each contributing unique strengths, mandates, and resources:

Stakeholder Category	Key Partners Involved	Anticipated Role in GMARCE
Higher Education & Research	Uganda Martyrs University, Muteesa I Royal University, technical institutes.	Leading action research, hosting the secretariat, designing training modules, providing technical expertise.
Local Government	District local councils (Masaka, Lwengo, Kalungu, Bukomansimbi, Rakai, Lyantonde, Kalangala, Ssembabule).	Aligning RCE activities with district development plans, enforcing environmental policies, mobilizing local communities.
Civil Society (NGOs/CBOs)	Regional environmental NGOs, farmers' cooperatives, youth groups, women's associations.	Implementing grassroots community projects, conducting public awareness campaigns, tracking localized environmental degradation.
Formal School System	Local primary and secondary schools, Teacher Training Institutions.	Activating school environmental clubs, adopting learner-centered ESD teaching methods, modeling school gardens.
Private Sector & Media	Local business associations, agricultural processors, regional radio stations (e.g., Buddu FM, Radio Maria).	Funding corporate social responsibility (CSR) eco-projects, broadcasting environmental education programs, adopting clean production.

12.0 Initial Action Plan and Core Activities (2014-2016)

Upon formal recognition, GMARCE will prioritize the execution of a foundational two-year action plan designed to build institutional visibility, establish baseline data, and implement pilot community interventions:

Activity 1: Baseline Assessment and Mapping of Local Indigenous Knowledge

Before rolling out extensive field programs, the RCE will commission a comprehensive baseline study across the constituent districts. This study will map localized hotspots of wetland and forest degradation, analyze current solid waste management roadblocks in urban councils, and systematically document disappearing indigenous knowledge regarding soil conservation and biodiversity preservation.

Activity 2: Teacher Training and Capacity Building in ESD

The Faculty of Education at Uganda Martyrs University will spearhead professional development workshops targeting secondary and primary school teachers. The training will shift focus away from purely theoretical environmental teaching toward practical, learner-centered ESD methodologies. Teachers will be trained on how to utilize school environments (such as setting up compound woodlots or compost units) as live teaching aids.

Activity 3: Launch of the "Greening Schools and Communities" Pilot Initiative

GMARCE will launch pilot projects in selected highly vulnerable areas. In rural areas, this will involve setting up community-managed tree nurseries focused on indigenous tree species and establishing demonstration gardens for climate-smart agroforestry. In urban and peri-urban centers, pilot initiatives will establish structured plastic sorting and organic waste composting schemes in collaboration with municipal markets.

Activity 4: Renewable Energy Demonstration and Outreach

To counter the massive over-reliance on wood fuel and charcoal, the RCE's energy working group will establish low-cost renewable energy demonstration centers. These centers will showcase operational biogas digesters constructed using locally available materials and promote the adoption of institutional fuel-efficient cookstoves in regional boarding schools and colleges, significantly reducing biomass consumption.

Activity 5: Public Advocacy and Community Media Campaigns

Leveraging the wide reach of local FM radio stations in the Greater Masaka region, the RCE will produce and broadcast weekly educational programs conducted in local languages. These sessions will break down complex environmental policies, discuss practical climate adaptation strategies for smallholder farmers, and provide a phone-in platform for community members to share ecological concerns.

13.0 Monitoring, Evaluation and Reporting

To keep activities aligned with core strategic objectives and ensure accountability to both local stakeholders and the Global RCE Network coordinated by the United Nations University Institute for the Advanced Study of Sustainability (UNU-IAS), a robust Monitoring and Evaluation (M&E) framework will be instituted.

13.1 Key Performance Indicators (KPIs)

The success of GMARCE interventions will be measured quantitatively and qualitatively through specific milestones, including:

- Number of school teachers formally trained and actively practicing ESD methodologies in their classrooms.
- Acreage of degraded wetlands or forest catchments restored through community-led tree planting campaigns.
- Number of active environmental clubs established or revitalized in regional schools.
- Quantity of organic waste successfully diverted from urban landfills into agricultural composting schemes.
- Number of collaborative research papers published focusing directly on addressing Greater Masaka's ecological challenges.

13.2 Reporting Mechanisms

The technical working groups will submit quarterly progress briefs to the Secretariat. The RCE Coordinator will synthesize these updates into an annual comprehensive GMARCE Progress Report. Once reviewed and approved by the Advisory Board, this report will be made accessible to all local network partners, published on the institutional portal, and shared officially with the Global RCE Service Centre.

14.0 Sustainability and Resource Mobilization

Achieving long-term viability requires a diversified financial and resource mobilization plan that does not rely solely on single-source funding. GMARCE will adopt a multi-pronged resource strategy:

14.1 Institutional Co-funding and In-kind Support

As the primary proponent and host, Uganda Martyrs University commits to providing long-term structural support. This includes office infrastructure, administrative overhead coverage, utilities, and allocating dedicated percentages of academic staff time to drive the research and coordination work of the Secretariat. Partner local governments will chip in by providing logistical support for field operations within their districts.

14.2 Grant Seeking and Project Proposals

The Secretariat, drawing on the academic expertise of UMU and partnered higher education institutions, will aggressively develop targeted project proposals. These proposals will target

international conservation funds, development agencies, and global climate funds (such as the Global Environment Facility or specialized ESD grants) to finance large-scale community research, clean energy installations, and agricultural extension training.

14.3 Community and Private Sector Co-production

Grassroots initiatives will build self-sustainability mechanisms into their designs. For instance, community-managed tree nurseries will generate modest income by selling high-yield fruit seedlings and exotic agroforestry varieties to commercial farmers, reinvesting profits into expanding indigenous species collections. Private sector partners, especially agro-processors, will be courted to sponsor specific environmental award schemes in schools as part of their marketing and sustainability mandates.

15.0 Conclusion

The establishment of the Greater Masaka Regional Centre of Expertise (GMARCE) represents a critical, timely milestone in turning the tide against environmental degradation and socio-economic vulnerability in central Uganda. By systematically breaking down the silos that separate academic institutions, local government administrators, civil society organizations, and local grassroots communities, the RCE constructs an integrated ecosystem of learning and action.

Through its firm commitment to Education for Sustainable Development (ESD), GMARCE will not only address immediate ecological emergencies like soil nutrient depletion, deforestation, and water contamination within the Lake Victoria basin, but it will also fundamentally reshape the mindset of the next generation. Equipping young learners, local farmers, and small-scale entrepreneurs with eco-literacy, practical sustainable tools, and an appreciation for ancestral ecological wisdom ensures the region's development remains deeply resilient.

Uganda Martyrs University, alongside its robust array of founding regional partners, stands fully prepared to provide the administrative structure, technical rigor, and community-driven passion required to transform the vision of GMARCE into a living, impactful reality. We look forward to formal recognition and integration into the global network of excellence.

These resources have attracted many development activities which have resulted into population increase, industrial development and rapid urbanization. Today, the catchment is the socio-economic hub of the region, housing the major urban centers. Over 80% of the regional population is employed in agriculture and related industries, and the catchment contributes a significant proportion to the National Gross Domestic Product (GDP).

However, the rapid population growth, high rate of urbanization, and industrialization have placed immense pressure on the natural resources of the region. This has manifested in terms of severe environmental degradation, pollution, biodiversity loss, and declining agricultural productivity.

The underlying causes of these challenges are complex and interrelated, ranging from weak

institutional capacities and policy implementation gaps to inadequate public awareness, limited technical skills, and widespread poverty. Addressing these challenges requires a paradigm shift from conventional, top-down approaches to integrated, multi-stakeholder and participatory strategies that actively engage local communities, academic institutions, and the private sector.

16.0 Operational Modalities and Working Framework

To translate the strategic objectives of GMARCE into tangible, field-level impacts, the Secretariat will institute a set of collaborative operational modalities. These modalities emphasize joint ownership, decentralized responsibility, and adaptive project management.

16.1 Regular Stakeholder Fora and Assembly

GMARCE will host a bi-annual Regional Sustainability Assembly. This platform will serve as a democratic forum where grassroots community groups, smallholder farmer representatives, school teachers, and local government technocrats can interact directly with university researchers and policy makers.

During these assemblies, progress reports from individual technical working groups will be presented, evaluated, and cross-referenced with community feedback. The assembly will also serve as a peer-review arena to assess the performance of the various pilot projects running across the constituent districts.

16.2 Shared Knowledge Repository and ICT Hub

Recognizing that information asymmetry hinders localized sustainable development, the Secretariat at Uganda Martyrs University will host an open-access Digital Sustainability Hub. This platform will curate, organize, and digitize:

- Localized environmental research papers and dissertations.
- Practical toolkits for agroforestry and soil regeneration tailored to regional soil types.
- Training manuals for teachers on active learning pedagogies for ESD.
- Audio recordings and scripts of community radio broadcasts for reuse by smaller community-based radio networks.

16.3 Collaborative Project Design and Co-financing

Whenever a localized intervention or grant opportunity is identified, GMARCE will not design proposals in isolation. The Secretariat will convene a cross-sectoral writing and planning team consisting of academic researchers (to ensure scientific validity), local NGO practitioners (to ensure cultural and community alignment), and district technical officers (to ensure alignment with national policy guidelines).

Co-financing mechanisms will be explicitly mapped out during the design phase, detailing the exact material, human, or financial inputs that each partner brings to the project table.

17.0 Strategic Alignment with National and International Frameworks

The establishment and operations of GMARCE do not occur in a vacuum; rather, they are designed to align closely with and actively accelerate the implementation of existing national development blueprints and global sustainability frameworks.

17.1 Alignment with Uganda Vision 2040 and NDP III

The Government of Uganda, through its Vision 2040 and the National Development Plan (NDP), places substantial emphasis on human capital development, sustainable agriculture, and natural resource management as key drivers of macroeconomic transformation.

GMARCE directly addresses these national priorities by building a highly skilled, eco-literate workforce through its teacher-training networks, and by introducing practical, low-cost soil and water conservation methodologies that bolster rural agricultural productivity and resilience against climate shocks.

17.2 Alignment with the Global Sustainable Development Agenda

The core tenets of GMARCE intersect directly with several United Nations Sustainable Development Goals (SDGs), most notably:

- **SDG 4 (Quality Education):** Specifically Target 4.7, which aims to ensure that all learners acquire the knowledge and skills needed to promote sustainable development.
- **SDG 13 (Climate Action):** By strengthening regional and local capacity for climate change adaptation, mitigation, and early warning systems.
- **SDG 15 (Life on Land):** Through active grassroots restoration of degraded forest ecosystems, wetlands, and the promotion of biodiversity preservation in the Lake Victoria basin.

18.0 Risk Management and Mitigation Strategy

Operating a large, decentralized multi-stakeholder network across eight districts presents distinct structural, political, and financial risks. GMARCE has proactively identified these risks and formulated corresponding mitigation strategies:

Identified Risk	Risk Level	Proactive Mitigation Strategy
Financial Instability: Over-reliance on erratic short-term grants leading to the stalling of core	High	Uganda Martyrs University provides baseline institutional co-funding (office space, staff time, utilities). Active

Secretariat coordination activities.		exploration of income-generating activities within pilot community projects.
Stakeholder Fatigue: Waning participation from partner local governments or civil society organizations over time due to slow initial project rollouts.	Medium	Maintaining high structural transparency, ensuring regular feedback loops through the bi-annual assembly, and celebrating quick, visible local wins in early pilot phases.
Political Interferences / Institutional Changes: Shifting political leadership or administrative restructuring within participating district local councils.	Low to Medium	Institutionalizing partnerships via formal Memoranda of Understanding (MoUs) signed with the technical offices (e.g., Chief Administrative Officers) rather than political figures.

19.0 Appendix A: Profile of the Host Institution (Uganda Martyrs University)

Uganda Martyrs University (UMU) is a private, co-educational Catholic university chartered by the Government of Uganda. Established in October 1993, the university has grown significantly in terms of student enrollment, academic programs, research outputs, and physical campuses spread across different regions of Uganda. UMU is deeply committed to achieving academic excellence while maintaining a strong orientation toward community transformation and social justice.

The Faculty of Education at UMU, which serves as the direct institutional anchor for the GMARCE Secretariat, possesses extensive experience in curriculum development, teacher professional development, and educational research. The faculty has been at the forefront of championing the integration of Education for Sustainable Development (ESD) into teacher-training curricula within the East African region.

Furthermore, UMU's main campus at Nkozi is located within the heart of the Greater Masaka region, providing an ideal geographical and logistical base for coordinating network activities, hosting stakeholder workshops, and managing community demonstration projects. The

university's established administrative systems, financial auditing frameworks, and ICT infrastructure guarantee a secure and transparent environment for managing RCE resources and partnerships.

20.0 Appendix B: Minutes of the Stakeholder Consultative Meeting

Date: 11th July 2013

Venue: Main Seminar Room, Uganda Martyrs University, Nkozi

Attendance: 42 participants (Representatives from local governments, schools, NGOs, and private sector).

Agenda:

1. Opening Remarks and Introduction to the RCE Concept.
2. Plenary Session: Identification of Regional Sustainability Challenges.
3. Group Work: Mapping Stakeholder Contributions and Action Planning.
4. Formation of the GMARCE Steering Committee.

Key Deliberations and Decisions:

- **Welcome Address:** The Dean of the Faculty of Education welcomed participants and highlighted the urgency of building a collaborative network to address the visible environmental degradation in the Greater Masaka region. He emphasized that the university cannot remain an ivory tower while surrounding communities face declining soil fertility and deforestation.
- **RCE Concept Presentation:** A detailed presentation was delivered on the global RCE network coordinated by the United Nations University (UNU). The presentation clarified that an RCE is not a new physical institution, but a strategic network to promote Education for Sustainable Development (ESD).
- **Problem Identification:** Participants actively shared field experiences regarding environmental challenges. Representatives from Kalangala highlighted the massive conversion of natural forests into commercial palm oil plantations, noting the loss of local biodiversity. Urban council officers from Masaka highlighted the severe financial constraints faced in managing municipal solid waste.
- **Action Planning:** The plenary agreed that the formal application should be drafted under five primary thematic areas: Ecology, Sustainable Agriculture, Energy, Health/Sanitation, and Formal Pedagogy. It was agreed that Uganda Martyrs University would take the lead in compiling the final draft document.
- **Steering Committee Selection:** A 7-member interim steering committee was elected to oversee the finalization of the application and coordinate initial stakeholder engagements. Bro. Byaruhanga Aloysius was chosen to chair the committee, with representatives drawn from Muteesa I Royal University, Masaka District Local Government, and a regional environmental NGO network.

Way Forward:

The chairperson thanked members for their commitment and stated that the draft application would be circulated to all key partners for review before final submission to the United Nations University Institute for the Advanced Study of Sustainability (UNU-IAS) in early 2014.

21.0 Appendix C: Letters of Support and Endorsement

(Note: This section contains a collection of official institutional endorsements confirming active partnership and formal backing for the establishment of the Greater Masaka Regional Centre of Expertise).

1. Endorsement from Masaka District Local Government

Signed by the Chief Administrative Officer (CAO), confirming the district's readiness to integrate GMARCE activities into the District Environment Action Plans (DEAPs) and provide field support for community pilot initiatives.

2. Endorsement from Muteesa I Royal University

Signed by the Vice Chancellor, affirming the institution's commitment to collaborative research, joint technical working group participation, and the co-hosting of regional sustainability seminars.

3. Endorsement from the Regional Centre of Expertise (RCE) Greater Pwani (Partner Peer RCE)

A letter of solidarity and mentorship from RCE Greater Pwani, offering technical guidance, institutional peer-review support, and cross-border knowledge exchange opportunities during the foundational phase of GMARCE.

4. Endorsement from the National Environment Management Authority (NEMA) – Regional Office

Signed by the Regional Environment Officer, certifying that the objectives of GMARCE are in perfect alignment with national environmental education strategies and pledging technical oversight support during wetland and forest restoration campaigns.

5. Endorsement from the Greater Masaka NGO Forum

Signed by the Executive Director, representing a coalition of over thirty local NGOs and CBOs, confirming their readiness to utilize their existing grassroots networks to implement GMARCE community literacy programs and pilot projects.